High-Performance Computing (HPC)
Used when datasets are too large for normal computers
It consist of
i. Clusters - collection of linked comps.
ii. GPUs - 1000s of cores for parallel math
iii. Supercomputers - extremely powerful

Cloud Computing
It models services includes
→ IaaS (Infrastructure as a Service) storage.
PaaS (Platform as a Server) tools for app development.
SaaS (Software as a Server) G-mails, Google Docs

: Benefits
a. Scalability
Pay-as-you-go
Collaboration.
Ease of backup

FUNDAMENTALS OF COMPUTER OPERATIONS
1) CPU architecture.
It is divided into.
a). Arithmetic Logic Unit (ALU)
Performs → Addition / Substraction
Logic operations
Comparision operation

STA 2102: INFORMATION TECHNOLOGY FOR STATISTICS

· Definition of IT
Study, design, development, implementation support or management of computer-base information system.

Core component
✓ Hardware - physical devices (CPU, disks)
✓ Softwares - operating systems, applications statistical programs
✓ Data - raw facts used for analysis
✓ Procedures - rules for using IT system
✓ People - Users, IT professionals, analyst

Data, information, knowledge.
✓ Data refers to raw unprocessed facts eg heart rate = 95 bpm
✓ Information are processed organized and interpreted data eg patient rec
Knowledge refers to application of information to make decision

Role of IT in modern Society
i) Health care and biostatistics
✓ Electronic Medical Records (EMRs)
✓ Clinical decision support systems
✓ Modeling of disease spread
✓ Big epidemiological databases
✓ Automated data collection tools
Government
✓ E-Governance, statistical reporting
Population census analytics

Input & output

COMPUTER HARDWARE I

1. Input Devices

Traditional → keyboard
Mouse
Joystick

Advanced:
- Image scanners
- Biometrics scanners
- Digital cameras
- Barcode readers
- Voice input systems!

2) Output Devices.

Softcopy output
- Monitor
- Projectors
- VR displays

Hardcopy output
- Printers
- Plotters (statistical graphs)

Measuring I/O performance
DPI (Dots per inch)
PPM (pages per minutes)

COMPUTER HARDWARE II

(storage & memory)

1 Primary Storage.

RAM → type of computer memory that stores data and instructions temporarily while the computer is running

features → Volatile Memory
→ Fast Access

Types of ~~RAM~~ · DRAM & SRAM

b). Control Unit (CU)

Directs operations of CPU
Manages execution of instruction

c) Registers

Small, fast memory inside CPU:
- Accumulator - temporary storage
- Program Counter - next instruction address
- Memory address (MAR)

Machine Cycle.

Every CPU instruction goes through
i). Fetch → get instruction from memory
Decode → interpret instruction
Excode → perform required operation
Store → Save the result

Fetch → Decode → Excode → Store.

Performance Measures.
- Clock speed

measured in GHz
- 1 GHz = 1 billion cycles per second

Cache memory
- L_1 (fastest)
- L_2
- L_3 (largest)

Parallel Processing

Used in big dat analysis, simulation methods, Time-consuming regressions neural networks & Real-time high frequency trading

COMPUTER SOFTWARES

1) System Software
Controls and manages the computer hardware. eg OS (operating systems) - Windows, Linux
functions - Manages hardware
- Runs and stops programs
- Handles files & storage

2) Application Software
designed to help users perform specific tasks.
G. purpose - Word processing (Msword)
- Spreadsheets (Excel
- Presentations (Power Point)
Specialized
· R & RStudio
Pythone
SPSS
STATA

Compilers Vs Interpreters
Com -> converts whole program - machine code
Interpreters - Executes lines by line.

Workflow Optimization:
Automation with macros
Scripting statistical procedures
Batch processing of dataset.

✓ROM - It is non volatile memory that stores data permanently.
features ✓Non volatile - Data remains even when is off
✓Holds BIOS & firmware
Very reliable.

✓Cache Memory - The fastest memory
- Improves CPU efficiency.

Secondary Storage.
HDD -> Hard disk Drive.
features -> Mechanical
- large capacity eg (1 TB, 2 TB etc
Cheaper

SSD - Solid State Drive - uses flash memory
- No moving parts
features - Fastest
✓lower power consumption

Optical Storage -> CDs, DVDs
Cloud Storage - Remote servers accessible via internet.

Disk Organization.
Disk has - Platters - round discs
- Surfaces - two sides of each plat
- Sector - small units of storage
- Clusters - group of sectors
✓Track - Concentric circles

Opening & editing files
Copying, moving & renaming
Deleting files
Backing up files
File compression
File security

File system

It controls how files are stored and organized on storage devices
eg NTFS – windows
FAT 32/ exFAT – flash drives
·ext4 – Linux

File lifecycle.
- ✓ Creation
- ✓ Editing
- ✓ Indexing
- ✓ Retrieval
- ✓ Backup
- ✓ Archiving.

Databases – is organized collection of data that is stored and managed so that it can be easily accessed.

key terms
- Table – contains rows & columns
- Record – (Row) single entry
- field – column
- Primary key – Unique identifier eg student ID
- Foreign key – field linking one table to another

DATA FILES & FILE MANAGEMENT

Data files is collection of related data stored on a computer, usually kept on storage devices like HDD

Types of data files
i) Text files – plain text characters eg txt csv
ii) Binary files, eg exe, dat,
iii) Structured data files – organized in rows /columns or key-value pairs eg xlsx
iv) Program files – software instructions eg exe, apk, .py, .jar.
v. Multimedia files – images audio video

Components of a file.
file Name
file Extension
file location
file Size

File management
Refers to process of organizing, naming storing, accessing, securing and controlling files.

Importance
Prevent loss
Save storage space
Ensures data security

Activities in file management
- ✓ Creation of file
- ✓ Naming file
- ✓ Saving & storing file